

FreelanceHub

A MERN freelance marketplace for posting work, hiring talent, chatting, holding escrow, cancelling with refunds, and completing jobs with reviews. Tests, CI, and Docker are part of the repo; Stripe, S3, Redis, and SMTP stay optional.

React 19 + Vite 6

Express + MongoDB

Socket.IO

httpOnly JWT

CI + Docker

Repository: github.com/ShadowAN-AB/Freelance_Marketplace

Local app: <http://localhost:5178> · **API:** <http://localhost:5001> · **Compose web:** <http://localhost:8080>

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1. What this project is

FreelanceHub is a full-stack marketplace where **clients** post paid projects and **freelancers** bid. After a hire, the two parties chat, the freelancer submits work (optionally with files), the client can request revisions, cancel with an escrow refund, or approve. Simulated escrow is released on complete. Both sides can then leave a review. An **admin** account can inspect users, projects, contracts/escrow, reports, and an audit log.

The product is a classic MERN monolith (MongoDB, Express, React, Node.js). The demo holds money as a `Payment` document: hire creates `held`, complete sets `released`, cancel sets `refunded`. Stripe, object storage, Redis, and SMTP are env-optional. If they are unset, the same hire loop still works.

One-line summary. Post → propose (or rebid after withdraw) → hire (escrow hold) → chat → submit work → optional revision *or* cancel/refund → complete → review.

2. Goals and non-goals

In scope

- Email/password auth with httpOnly cookies, refresh tokens, and Bearer JWT
- Email verify and password reset (JSON mailer or SMTP)
- Project posting, URL-backed search, skill matching
- One pending proposal per freelancer per project; rebid after withdraw
- Hire, contract, escrow hold / release / refund
- Deliverable uploads (disk or optional S3)
- Project-scoped chat with join auth, multi-tab presence, typing
- Saved items, shortlists, reports, verified-skill tags
- Public client pages and richer freelancer portfolios
- Admin analytics, contracts/escrow, audit log (no impersonation)
- API tests, GitHub Actions CI, lint, Docker Compose

Out of scope (still)

- Google / social login
- LLM matching or generated cover letters
- Milestones / partial releases
- Native mobile apps
- Multi-currency or a microservice split
- Mandatory Stripe — it is a vertical slice behind env vars, not the default demo

3. Technology stack

Layer	Choice	Why it is used
Client	React 19, Vite 6, Tailwind CSS 4	SPA UI, fast local dev, utility styling
Routing / data	React Router 7, TanStack Query, Axios	URL filters, cached REST, <code>withCredentials</code>
Realtime client	Socket.IO client	Messages, presence, typing
Charts	Recharts	Admin analytics
API	Node.js, Express 4	REST + static <code>/uploads</code>
Data	MongoDB 7, Mongoose 8	Users, projects, contracts, audit log
Auth	JWT access + refresh, bcryptjs, httpOnly cookies	Session cookies plus Bearer for sockets
Validation	Zod	Request body schemas
Uploads	Multer + optional S3	Avatars 2 MB; deliverables 8 MB
Realtime server	Socket.IO + optional Redis adapter	Rooms, presence sets, multi-process
Mail	Nodemailer	JSON transport in demo; SMTP when set
Quality	node:test + supertest, Vitest, ESLint, Prettier	Hire-loop API tests and client helpers
Hardening	Helmet, CORS, request ids, rate limits	Baseline HTTP headers and logs

Local ports avoid macOS conflicts: API **5001**, Vite **5178**. Vite proxies `/api`, `/uploads`, `/health`, and `/socket.io` to the API. `CLIENT_URL` is required; the process will not start without it.

4. System architecture

The browser never talks to MongoDB. It talks to Express over HTTP (JSON and multipart, cookies included) and to the same Node process over WebSockets. Access tokens travel as httpOnly cookie `fh_access` and/or `Authorization: Bearer`. Refresh uses cookie `fh_refresh` (or a JSON body). Socket.IO authenticates the handshake token or the access cookie.

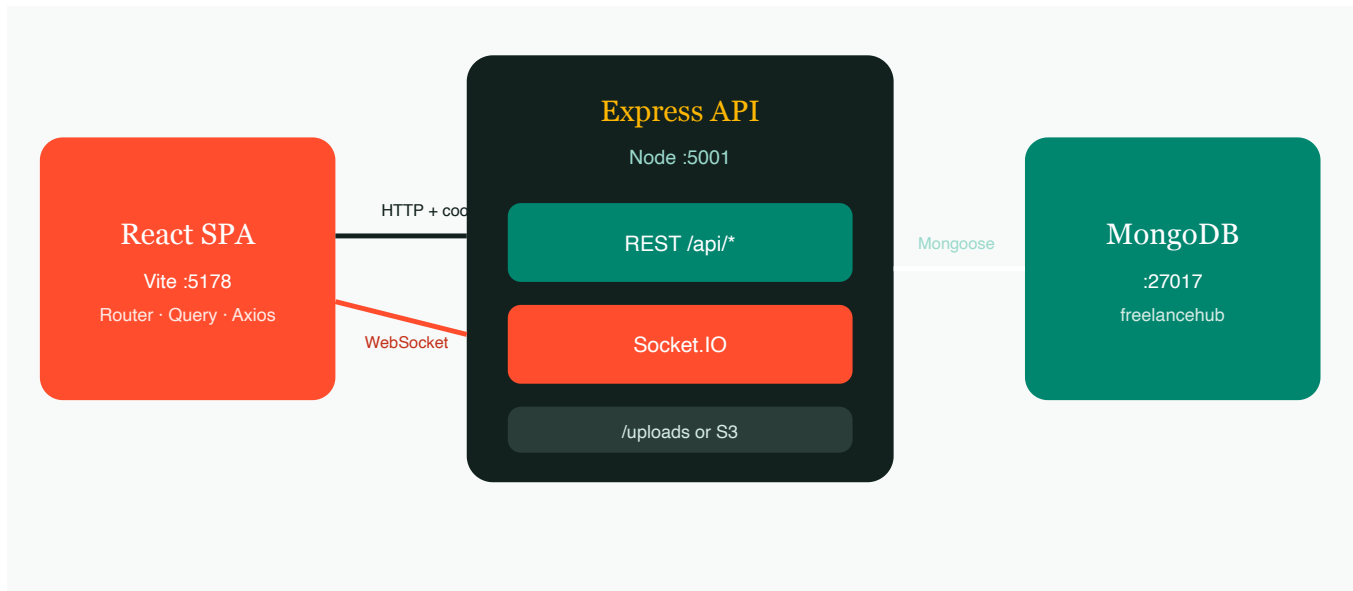


Figure 1. Browser SPA, single Express process (REST + Socket.IO + files), MongoDB. Optional Redis/S3/SMTP/Stripe attach via env.

On load, AuthContext calls `GET /api/auth/me` (cookies or stored Bearer token). A 401 triggers `POST /api/auth/refresh`; if that fails the UI signs out. Toasts replace `alert()` for proposal success and work actions.

5. Repository layout

```
Freelance_Marketplace/
├── client/                    React SPA (Vite)
│   ├── src/App.jsx           Routes, ToastProvider, role guards
│   ├── src/context/          Auth, toasts, 401 refresh
│   ├── src/layouts/          Public nav and signed-in shell
│   ├── src/pages/            public, auth, client, freelancer, admin, shared
│   ├── Dockerfile            nginx + /api proxy
│   └── vite.config.js         Proxies → :5001 · Vitest
├── server/
│   ├── src/server.js          createApp() + start(); health checks Mongo
│   ├── src/models/            Including AuditLog
│   ├── src/controllers/       Route handlers
│   ├── src/routes/            Express routers
│   ├── src/middleware/        JWT/cookies, Zod, Multer
│   ├── src/services/          socket, mailer, storage, payments, audit, logger
│   ├── src/seeds/seed.js      Demo data (refuses NODE_ENV=production)
│   ├── test/api.test.js       Hire-loop supertest suite
│   └── uploads/               Local files when S3 is unset
├── .github/workflows/ci.yml
├── docker-compose.yml         mongo + api + web
├── docs/                      This HTML + PDF
└── README.md
```

A native `mongod` on port 27017 can hide an empty Docker volume. The API uses whatever is on `MONGO_URI`.

6. Domain model

Collections are linked by ObjectIds. A project has at most one hired proposal. A contract is unique per project. A payment is unique per contract. Chat is unique per (project, unordered pair of users). Admin actions write `AuditLog` rows (blocks, report updates, skill verification). Impersonation is disabled.

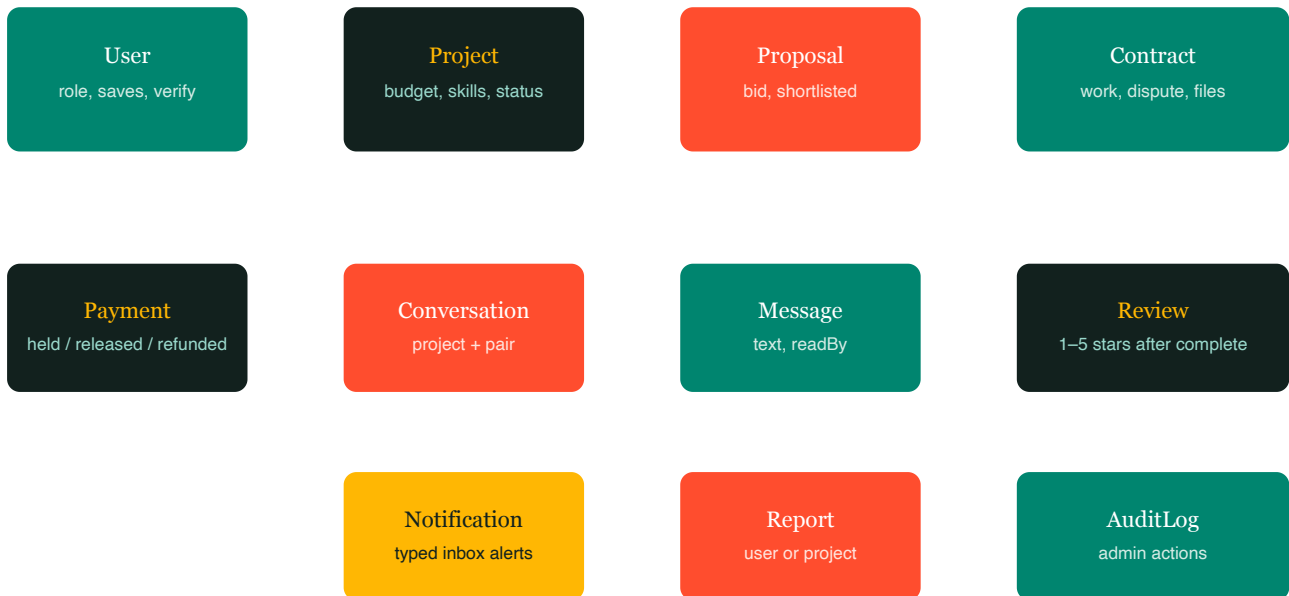


Figure 2. Main collections plus audit log. Users also hold saved ids, refresh-token hash, and verified skills.

Important statuses

Entity	Statuses
User.role	freelancer , client , admin
Project	open → in_progress → completed (or cancelled)
Proposal	pending , accepted , rejected , withdrawn (withdrawn may become pending again)
Contract	active , completed , cancelled (stores disputeReason)
Payment	held , released , refunded (simulated or stripe)
Report	open , reviewed , dismissed

7. Roles and screens

Role	Can do	Primary screens
Guest	Browse projects and talent, open public client pages. Must log in to bid, save, report, or chat.	Home, /projects, /freelancers, /clients/:id, login, register, forgot/reset password
Freelancer	Propose on open projects (form hides after a pending/accepted bid; rebid after withdraw), save projects, submit work, cancel an active contract, chat after a proposal exists, review after completion.	Dashboard, Browse, Proposals, Active work, Saved, Messages, Earnings, Profile, Settings
Client	Post/edit/cancel open projects, shortlist and accept/reject, save talent, request revision, approve (release), or cancel/refund, report, chat after a proposal exists.	Dashboard, Post project, My projects, Proposals inbox, Active work, Saved, Messages, Profile, Settings
Admin	Stats, block users, verify freelancer skills, list projects, open contracts/escrow, read the audit log, resolve reports. Seeded only — not self-registered.	/admin, Users, Projects, Reports, Contracts, Audit

8. Core hiring loop

This is the money path. Saved items, shortlists, reports, and presence sit on top of it.

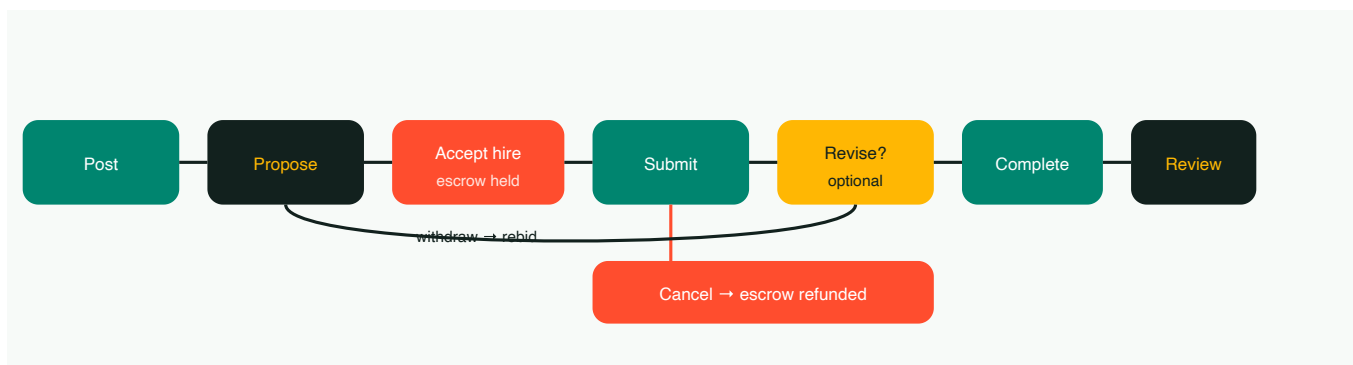


Figure 3. Happy path, revision loop, withdraw/rebid, and cancel/refund. Complete still requires a current `workSubmittedAt`.

Domain rules (enforced on the server)

- Unique index on `{ projectId, freelancerId }`. A second pending bid is 409. A `withdrawn` row is updated back to `pending`.
- You cannot bid on your own project or on a non-open project. After a proposal exists, the project detail page shows status instead of a second form.
- Accepting a bid rejects other pending bids, sets the project to `in_progress`, creates a contract and a held payment, and opens a conversation.

- Chat requires a pending or accepted proposal. Socket `conversation:join` is participant-checked (join is not an open room id).
- Only the hired freelancer can submit work. Only the client can request a revision or complete.
- Either party (or admin) can cancel an `active` contract with a reason; held escrow becomes `refunded` and the project is cancelled.
- Revision keeps status `active`, increments `revisionCount`, stores `revisionNote`, and clears `workSubmittedAt`.
- Reviews are allowed only after `completed`, once per reviewer per contract.

9. Feature catalogue

9.1 Auth and profiles

Register is limited to `freelancer` or `client`. Passwords are hashed with bcrypt (cost 12). Login is rate-limited (40 / 15 minutes). Sessions set httpOnly cookies and return a Bearer token for sockets. `POST /auth/refresh` rotates refresh hashes. `POST /auth/logout` clears cookies even if the access token already expired. Email verification and password reset send mail through Nodemailer (JSON log in demo). `SKIP_EMAIL_VERIFY=true` (and `NODE_ENV=test`) skip the verify gate for local/seeded accounts.

Profiles store bio, location, avatar, freelancer skills/rate/portfolio images/availability, verified-skill tags (admin-set), or client company name. Public client pages list that company's projects.

9.2 Projects and search

Browse keeps `q`, category, due-soon, and page in the URL. Talent browse keeps hourly-rate bounds in the URL. Cards show the client company. Matching score is skill overlap, capped at 100%.

9.3 Proposals and shortlist

Cover letter, bid amount, estimated days. Clients star pending proposals. Inbox sorts shortlisted first.

9.4 Work, deliverables, revisions, disputes

Submit-work is multipart (PDF, ZIP, PNG, JPEG, WebP, 8 MB, up to 8 files). Files persist to disk or S3. The work card can Approve, Request revision, or Cancel & refund escrow.

9.5 Saved items

Freelancers bookmark projects; clients bookmark talent. `/app/saved` lists them.

9.6 Messaging, unread, presence, typing

Unread count is threads whose latest message is not in `readBy`. Presence is a **set of socket ids per user** so a second tab does not kick the first offline. Chat notifications fire only if the other party is offline. Typing events stay inside the conversation room.

9.7 Notifications

Proposal received/accepted/rejected, work submitted, revision requested, contract completed, contract cancelled, review received, report updates. Alerts use a keyboard-focusable disclosure and `aria-live` unread counts. Success toasts use `aria-live="polite"`.

9.8 Reports and admin

Users report a user or project. Admins review or dismiss (audited). Admins block users (audited), verify skills (audited), and open contracts with escrow status. There is no impersonate control.

10. Client application

Path	Who	Page
<code>/</code>	Public	Marketing home
<code>/projects</code> , <code>/projects/:id</code>	Public	URL filters; bid form or “your proposal” status
<code>/freelancers</code> , <code>/freelancers/:id</code>	Public	Hourly-rate filters, verified skills, portfolio images
<code>/clients/:id</code>	Public	Client company page and their projects
<code>/login</code> , <code>/register</code> , <code>/forgot-password</code> , <code>/reset-password</code> , <code>/verify-email</code>	Guest	Auth and recovery
<code>/app/work</code>	Both	Files, revision, approve, cancel/refund, review
<code>/app/messages</code>	Both	Threads, presence, typing
<code>/admin/contracts</code> , <code>/admin/audit</code>	Admin	Escrow list and action log

Visual language is unchanged: mint paper, coral buttons, teal accents, saffron highlights. Nested dashboard routes render through an `Outlet` in `AppShell`.

11. REST API

Base URL in the browser is `/api` (proxied). JSON errors: `{ message, errors? }`. List endpoints for contracts and conversations are paginated (`page` , `limit` , `pages` , `total`). Messages already paginated.

Method	Path	Auth	Purpose
POST	/auth/register, /auth/login	Public (limited)	Cookies + JWT
POST	/auth/refresh, /auth/logout	Cookie / optional	Rotate or clear session
GET	/auth/me	User	Current user
GET / POST	/auth/verify-email, /forgot-password, /reset-password	Public	Email flows
PATCH	/auth/password	User	Change password
GET	/projects/:id	Optional	Project + <code>myProposal</code> for the bidder
POST	/contracts/:id/cancel	Party	<code>{ reason }</code> → refund held escrow
GET	/payments/me, /payments/mode	User	Ledger; <code>simulated</code> vs <code>stripe</code>
POST	/payments/webhook	Stripe signature	Ignored unless secrets are set
GET	/admin/contracts, /admin/audit	Admin	Escrow and audit log
PATCH	/admin/users/:id/verify-skills	Admin	Verified skill tags
GET	/health	Public	<code>{ ok, mongo, service }</code> — 503 if Mongo is down

Earlier catalogue routes (proposals, submit-work, revision, complete, conversations, notifications, reports) are unchanged in purpose; see the source routers for the full list.

12. Real-time chat and presence

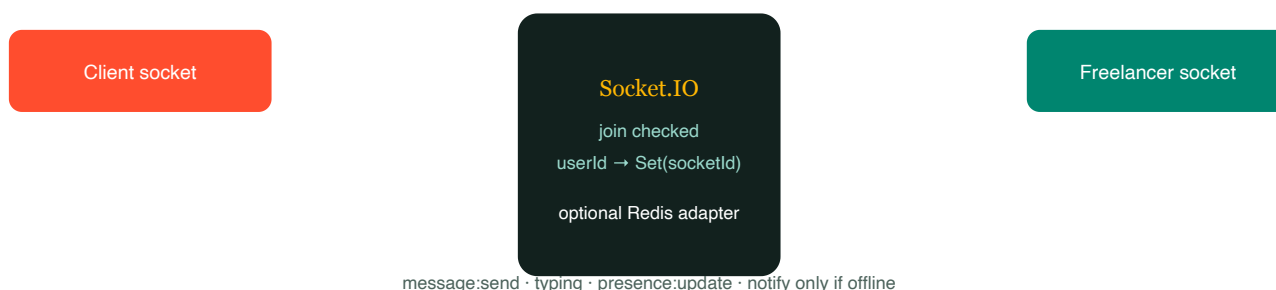


Figure 4. Join is authorized. Presence is a set of sockets per user. Chat alerts are quiet when the other party is online.

Event	Direction	Meaning
<code>conversation:join</code>	Client → server	Join only if the user is a participant
<code>typing</code>	Both	In-room typing indicator
<code>message:send</code>	Client → server	Persist, broadcast; notify if peer offline
<code>message:new</code> / <code>message:read</code>	Server → room	Live thread and receipts
<code>presence:update</code>	Server → all	Online only when the last tab disconnects

13. Security, auth, and validation

- Passwords bcrypt-hashed; secret fields stripped from public user payloads.
- Access cookie `fh_access` (httpOnly, SameSite lax, Secure in production) plus refresh cookie `fh_refresh`. Refresh token is stored hashed on the user.
- Axios sends cookies and Bearer; 401 retries refresh once, then clears the UI session.
- CORS origin is exactly `CLIENT_URL` with credentials. Socket.IO uses the same origin.
- Zod on mutating routes. Multer rejects disallowed types. JSON body 1 MB. Auth rate limits skipped only in tests.
- Request id header `X-Request-Id` on every response. Structured JSON logs.
- Controllers still enforce ownership (hired party, conversation participant, and so on).

Demo secrets. Rotate `JWT_SECRET` before any shared deploy. Disk `server/uploads/` does not survive multiple API instances — set S3 for that. Payments are simulated unless Stripe env vars are set.

14. Tests, CI, and quality

`cd server && npm test` runs node:test + supertest against `freelancehub_test` (or `MONGO_URI_TEST`). Coverage includes register/login, unique proposal + rebid after withdraw, accept → contract + held payment, complete blocked without submission, revision clearing `workSubmittedAt`, chat 403 for outsiders, cancel → refunded, and `myProposal` on project detail.

`cd client && npm test` runs Vitest on format helpers. ESLint and Prettier configs live in each package. GitHub Actions on `main` starts Mongo 7, runs server lint/test, then client lint/test/build.

15. Local setup and demo

Prerequisites

- Node.js 20+ (tested on 22)

- MongoDB on `127.0.0.1:27017` — Docker `mongo` service or local `mongod`

```
cp .env.example server/.env
cd server && npm install && npm run seed && npm run dev
cd client && npm install && npm run dev
```

Open <http://localhost:5178>. Health: <http://localhost:5001/health> (includes `mongo: true`).

```
JWT_SECRET=replace-with-a-long-random-string docker compose up --build
```

Compose serves nginx on <http://localhost:8080> and proxies API/socket/uploads. Do not run `npm run seed` in production — the seed script exits if `NODE_ENV=production`.

Every seeded account uses password `Password123!`.

Role	Email	Notes
Admin	admin@freelancehub.dev	Users, contracts, audit, reports
Client	priya@freelancehub.dev	Open projects + in-progress inventory hire with Aisha
Client	arjun@freelancehub.dev	Completed SEO contract with Leo
Freelancer	aisha@freelancehub.dev	Pending dashboard bid + active mobile contract; verified React/Node tags after a fresh seed
Freelancer	kabir@..., meera@..., leo@...	Extra bids and a completed job

Ten-minute walkthrough

1. Log in as Aisha. Open the logistics dashboard project — you should see **Your proposal / Pending**, not a second form.
2. Log in as Priya. Work card: download deliverables, request revision, or Cancel & refund (do not cancel the seeded hire if you still want that demo path). Approve still releases escrow after a current submission.
3. Withdraw a pending bid as a freelancer, then rebid on the same project.
4. Browse `/projects?q=react` and `/freelancers?minRate=1500` — filters stay in the URL. Company names show on project cards.
5. Open Messages on two accounts: presence dots, typing line, quieter alerts when the other tab is online.
6. As admin, open Contracts (escrow column) and Audit after a block or skill verify.

16. Operations and production hooks

Concern	Demo default	Production hook
Auth	Cookies + Bearer, 7d access in <code>.env</code>	Shorter <code>JWT_EXPIRES_IN</code> , unique <code>JWT_SECRET</code>
Mail	Nodemailer JSON transport (logged)	<code>SMTP_URL</code> , <code>MAIL_FROM</code>
Files	<code>server/uploads/</code>	<code>AWS_S3_BUCKET</code> (+ region / public URL)
Presence	In-memory socket id sets	<code>REDIS_URL</code> → Socket.IO Redis adapter
Payments	<code>Payment.status</code> flipped in complete/cancel	<code>STRIPE_SECRET_KEY</code> + webhook secret
Health	<code>/health</code> 200/503 from Mongo readyState	Load balancer on that URL

- Seed wipes marketplace collections. Never point it at a live database.
- Vite 6 is pinned because newer Vite/Rolldown builds failed on this machine's native bindings.
- If port 27017 already has a local MongoDB, Docker's mapped port will fail or you will see an empty database. Point `MONGO_URI` at the instance that contains the seed.

Source of truth. This document describes the codebase as of 21 September 2026 on `main` , including the sequential feature pack (saved items, revisions, deliverables, shortlists, unread badges, settings/reports, presence) and the improvement slices (UX/session/sockets, tests/CI, cancel/rebid/search, cookie auth/Docker/optional adapters).